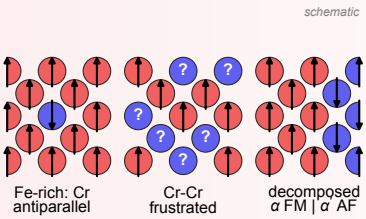
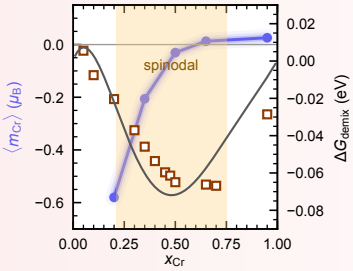
FRUSTRATE

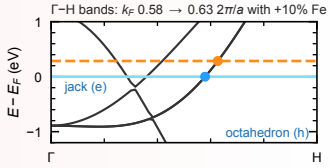
(a) atoms & moments

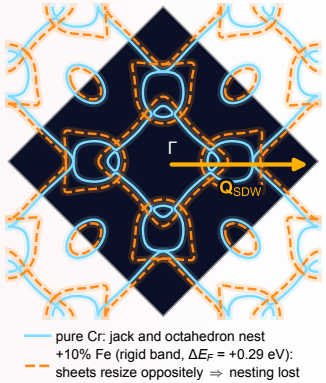




NEST

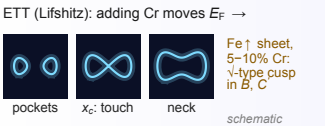
(b) Cr Fermi surface vs Fe content

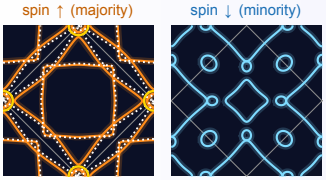




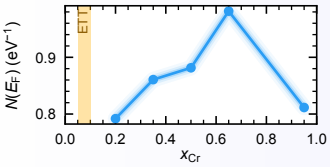
TOPOLOGY

(c) Fe-rich ETT: spin ↑ vs ↓



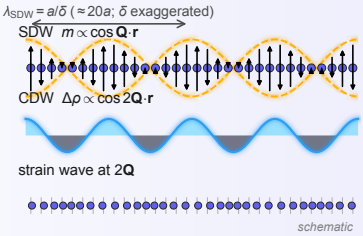


Cr shifts the majority states (band filling alone moves E_F away) ⇒ ETT at 5–10% Cr in spin ↑ only (Korzhavyi 2009); unfolded SQS Fermi surfaces pending

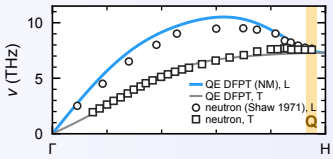


MODULATE

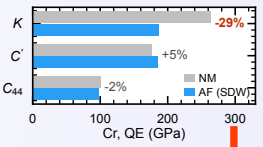
(d) SDW softens *K*, not *G*



Kohn dip near **Q**: zone-boundary L phonons only; C_{ij} = slopes at Γ (NM too steep vs AF neutron data)



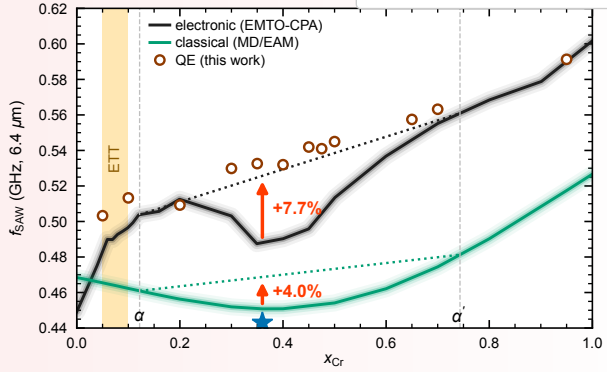
AF order: *K* −29%, shear unchanged



STIFFEN

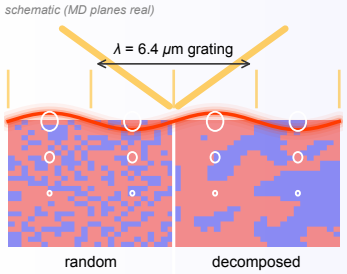
(e) electron → lattice: decomposition

$C = C_{ion} + C_{band}$, $C_{band} \propto -\sum_k (\partial\varepsilon_k/\partial\eta)^2\,\delta(\varepsilon_k - E_F)$



PROPAGATE

(f) Rayleigh wave in TGS



$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$

Fe−36Cr, MD: 451 → 473 MHz (+5.0%)

electronic lever rule: +7.7%

DETECT

(g) (P)SAW vs pristine CASS

